

Integrated attitude determination and control system via magnetic measurements and actuation

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ABSTRACT

A nonlinear control scheme using a Modified State-Dependent Riccati Equation (MSDRE) is developed through a pseudo-linearization of spacecraft augmented nonlinear dynamics and kinematics. The full-state knowledge required for the control loop is provided through a generalized algorithm for spacecraft three-axis attitude and rate estimation based on the utilization of magnetometer measurements and their time derivatives, while the control torque is generated via magnetorquers. The stability of the controller is investigated through Lyapunov function analysis and the local observability of the estimator is verified. The resulted attitude determination and control system has shown the capability of estimating the attitude better than 5 deg and rate of order 0.03 deg/s in addition to maintain the pointing accuracy within 5 deg in each axis with pointing stability of less than 0.05 deg/s. Monte-Carlo simulations are used to demonstrate the global asymptotic stability of the controller and the estimator for various initial conditions.

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1. Introduction

The aim of this paper is to develop a low cost three-axis Attitude Determination and Control System (ADCS) to fulfill spacecraft mission requirements during low accuracy modes which are the most frequent modes during mission life time. Many previous efforts extensively handled the problem of attitude determination and stabilization based on using magnetometer measurements and electromagnetic torquing separately. Previous researches on the State-Dependent Riccati Equation (SDRE) were done by Pearson [1], Garrard et al. [2], Burghart [3], and Wernli and Cook [4]. Recently, the SDRE control has been studied by Krikelis and Kiriakidis [5], and Cloutier et al. [6,7] with the application of the SDRE to a nonlinear benchmark problem in Mracek and Cloutier [8] and Chang et al. [9,10]. Hammitt et al. introduced controllability issues for the SDRE in Ref. [11]. A comparison study

involving the SDRE among other nonlinear control method was done in Ref. [12], and tracking control and state estimation methods using the SDRE were developed in Ref. [13]. Periodic Linear Quadratic Regulators (LQR) already has been developed for slightly perturbed spacecraft stabilization about nominal attitude via magnetic torquers. The linearized spacecraft dynamics is treated as a periodic system since such system is controllable if the orbit is inclined because the Earth's magnetic field vector rotates in space as the spacecraft moves around its orbit. Then the controller is obtained based on a transient or steady-state solution of the Riccati equation while the magnetic field is modeled as dipole or averaged over a time interval corresponding to a common period for the satellite revolution about the Earth and the Earth's own rotation [14, 15]. Also, Marco Lovera et al. [16] proposed a solution to spacecraft attitude control using magnetic actuation in terms of classical LQ periodic optimal control and extensions thereof, aiming at achieving efficient rejection of periodic disturbance torques was presented. The approach introduced by Challa et al. [17, 18], for attitude and rate estimation using only magnetometer measurements, uses a

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deterministic batch algorithm in conjunction with extended Kalman filtering. Crassidis and Markley [19] derived a real-time predictive filter for spacecraft attitude estimation without utilizing gyro measurements. The filter is based on a predictive tracking algorithm developed by Lu [20], which basically deals with control problems. Psiaki [21] developed a self-initializing filter to estimate three-axis attitude and rate from magnetometer data only. It uses a new attitude parameterization that is based on perturbations to the minimum quaternion that aligns the magnetic field in reference coordinates with the field in spacecraft coordinates. The filter model includes Euler dynamics and attitude kinematics, and its least-squares cost function penalizes magnetometer measurement errors and torque process noise.

The current work introduces a generalized nonlinear control scheme using a Modified State-Dependent Riccati Equation (MSDRE). The system dynamics equation is represented by the spacecraft nonlinear dynamics with momentum bias including gravity gradient, aerodynamic, and magnetic residual torques. Then, the quaternion kinematics is augmented with spacecraft dynamics to represent the overall process dynamics. A pseudo-linear formulation of the augmented system is developed while a MSDRE controller derived to solve a trajectory-tracking/model-following problem. The derivatives of the state dependent matrix w.r.t the states are taken into account through the development of the final MSDRE controller such that an optimal control signal is obtained rather than producing a suboptimal controller when these derivatives are ignored. To obtain the optimum control signal full-state knowledge is required, so attitude and rate filter is integrated to the control loop. The proposed filter is previously developed by the authors for spacecraft three-axis attitude and rate estimation utilizing magnetometer measurements only [22]. The architecture of the filter follows the extended Kalman filter EKF algorithm. Its structure is built up using spacecraft nonlinear dynamics and kinematics augmentation as in the MSDRE controller. Due to the interference between the magnetometer and magnetorquers which are used as main actuators, satellite dipole moment is added to the state vector. Also, the aerodynamic drag coefficient is included in the state vector for more filter robustness. Magnetometer measurements and its corresponding time derivatives are used to represent the filter measurement model. Except for the local linearization of the EKF scheme, the filter avoids any additional approximations regarding the attitude parameters and dynamics. Hence, analytical closed form solutions for the derivatives necessary to calculate the Jacobian matrix through the EKF algorithm is derived. The global asymptotic stability of the controller is investigated by applying Lyapunov theorem, and concluded by introducing stability regimes for the overall system which verify the stability conditions of the controller and the required accuracy of the filter. To test the developed ADCS, EgyptSat-1, launched in the last April, is used as a real test case where a magnetometer and fiber optic gyros are used for attitude/rate determination and magnetorquers/reaction wheels for stabilization. Basically, the ADCS has been tested to control the satellite and estimate the attitude and rates during detumbling and standby

modes where the magnetometer measurements and magnetorquers only can fulfill the mission attitude and rate accuracy requirements for these two modes. Except for the imaging (high accuracy) mode, the developed ADCS works as the main controller and attitude/rate estimator algorithm and not as a contingency algorithm in case of actuator/sensor failure which was the essential focus of many former research works. Finally, magnetometer measurements and magnetorquers are used through this paper only due to the hardware configuration of the test case in hand, but any other actuators or reference vector measurements such as Sun vector can be used.

The main contributions of this paper can be summarized first, in the development of a Modified State-Dependent Riccati Equation (MSDRE) controller which includes the derivatives of the state matrix such that an optimal control signal is obtained rather than producing suboptimal control signal when these derivatives are ignored. Second the derivation of a new pseudo-linear formulation of the augmented spacecraft non-linear dynamics and kinematics. The formulation includes the gravity gradient, aerodynamics, and magnetic residual torques in a moment biased system. The MSDRE controller is adopted to solve a trajectory-tracking/model-following problem and not only to solve a regulation problem. Third, a new previously developed filter for attitude and rate estimation using magnetic field measurements and their derivatives only is introduced and integrated with the MSDRE controller to represent a new developed ADCS based on using magnetic field measurements and actuation only. Fourth, the global asymptotic stability of the controller is investigated by applying Lyapunov function analysis and the local observability of the estimator is verified. Finally, the developed ADCS is applied to a real test case.

The next section describes briefly the statement of the problem, and then the MSDRE controller is developed and adopted for tracking problem. Through the fourth section the pseudo-linear representation of the system is derived and the stability issues are introduced. Before the simulation section, the state estimator design appears and is discussed. Then, the paper is concluded by testing the proposed ADCS via simulations and real flight data.

2. Statement of the problem

Consider the nonlinear system

$$\dot{x} = f(x, t) + v_x(t) \quad (1)$$

we will try to control this system such that its output is expressed as

$$y = cx \quad (2)$$

with the availability of a set of measurements

$$z = h(x, t) + v_z(t) \quad (3)$$

where $v_x(t)$ and $v_z(t)$ are random zero-mean Gaussian white-noise processes described

$$E\{v_x(t)v_x^T(t)\} = Q_f\delta(t-\tau) \quad (4)$$

and

$$E\{v_z(t)v_z^T(t)\} = R_f \delta(t-\tau) \quad (5)$$

where $\delta(t-\tau)$ is the Dirac delta function, E denotes expectation, while Q_f and R_f represent process and measurement noise matrices, respectively.

Now the task is to design a controller such that the system tracks a specified reference trajectory while a state estimator should be plugged in to provide the necessary states to drive the controller. The reference trajectory is given by

$$\dot{\chi} = F\chi \quad (6)$$

$$y_{ref} = C\chi \quad (7)$$

where the pair $[F, C]$ is supposed to be completely observable.

3. The MSDRE controller design

The control approach will be based on using the Differential State-Dependent Riccati Equation (DSDRE) to solve trajectory-tracking/model-following problem as follows:

Let us start by converting the nonlinear dynamics into a pseudo-linear system

$$\dot{x} = A(x)x + Gu \quad (8)$$

where u is the control signal. Note the process noise part is dropped in the controller design and it will be included through the estimator development.

A new dynamic system can be derived by augmenting the pseudo-linear system with the reference trajectory model such that

$$\dot{\tilde{x}} = \tilde{A}(\tilde{x}, t)\tilde{x} + \tilde{G}u \quad (9)$$

where

$$\tilde{x} = \begin{bmatrix} x \\ \chi \end{bmatrix}, \quad \tilde{A} = \begin{bmatrix} A & 0 \\ 0 & F \end{bmatrix} \quad \text{and} \quad \tilde{G} = \begin{bmatrix} G \\ 0 \end{bmatrix}$$

Then a quadratic cost function J is defined as

$$J = \frac{1}{2} \int_{t_0}^{t_f} [\tilde{x}^T \tilde{Q} \tilde{x} + u^T R u] dt \quad (10)$$

where

$$\tilde{Q} = \begin{bmatrix} Q & -QC \\ -C^T Q & C^T Q C \end{bmatrix} \quad (11)$$

and Q is nonnegative definite symmetric and R is positive definite symmetric.

The Hamiltonian of the system is given by

$$H(\tilde{x}, p, u) = p^T (\tilde{A}(\tilde{x})\tilde{x} + \tilde{G}u) + \frac{1}{2} (\tilde{x}^T \tilde{Q} \tilde{x} + u^T R u) \quad (12)$$

By satisfying the necessary conditions for optimality, a generalized Riccati equation can be derived as

$$\begin{aligned} -\dot{\tilde{P}} = & [\tilde{P}\tilde{A} + \tilde{A}^T \tilde{P} - \tilde{P}\tilde{G}R^{-1}\tilde{G}^T \tilde{P} + \tilde{Q}] \\ & + \left[\left(\frac{\partial \tilde{A}}{\partial \tilde{x}_i} \tilde{x} \right)^T \tilde{P} + \left(\frac{\partial \tilde{G}}{\partial \tilde{x}_i} u \right)^T \tilde{P} - \tilde{P}\tilde{G}R^{-1} \left(\frac{\partial \tilde{G}}{\partial u} u \right)^T \tilde{P} \right] \end{aligned} \quad (13)$$

where $\tilde{P}(t)$ is a positive definite matrix defined such that

$$p(t) = \tilde{P}(t)\tilde{x}(t) \quad (14)$$

The first part of Eq. (13) is the well-known state-dependent Riccati equation, and the second part is due to the nonlinearity of the system. Neglecting the second part leads only to a suboptimal control. Through this work the matrix $\tilde{G}$ is only a function of time, so the Modified State-Dependent Riccati Equation MSDRE for optimal control is

$$-\dot{\tilde{P}} = \tilde{P}\tilde{A} + \tilde{A}^T \tilde{P} - \tilde{P}\tilde{G}R^{-1}\tilde{G}^T \tilde{P} + \left(\frac{\partial \tilde{A}}{\partial \tilde{x}_i} \tilde{x} \right)^T \tilde{P} + \tilde{Q} \quad (15)$$

with a boundary condition

$$\tilde{P}(t_f) = 0 \quad (16)$$

gives immediately the optimal control u

$$u = -R^{-1}\tilde{G}^T \tilde{P}\tilde{x} \quad (17)$$

The reference trajectory y_{ref} is usually given in a form of points calculated offline and can be interpolated with time so, let us try to avoid the pair $[F, C]$ in the Riccati equation.

The matrix $\tilde{P}$ can be partitioned as

$$\tilde{P} = \begin{bmatrix} P & P_{12} \\ P_{21} & P_{22} \end{bmatrix} \quad (18)$$

Define a new vector b_{ext} , where ext stands for an external control signal as we will see

$$b_{ext} = P_{12}\chi \quad (19)$$

Using reference trajectory model in Eqs. (6) and (7) with the differential matrix equation for P_{12} which can be derived from Eq. (15) leads to

$$-\dot{b}_{ext} = \left[A^T + \left(\frac{\partial A}{\partial x_i} x \right)^T - PGR^{-1}G^T \right] b_{ext} - Qy_{ref} \quad (20)$$

Similar treatment can be applied for P_{21} and P_{22} by defining two new vectors c_{ext} and d_{ext} as follows:

$$c_{ext} = P_{21}^T \chi \quad (21)$$

and

$$d_{ext} = \chi^T P_{22} \chi \quad (22)$$

Then the differential equations of P_{21} and P_{22} can be replaced by

$$-\dot{c}_{ext} = [A - GR^{-1}G^T P]^T c_{ext} - Qy_{ref} \quad (23)$$

and

$$\dot{d}_{ext} = c_{ext}^T GR^{-1}G^T b_{ext} - y_{ref}^T Qy_{ref} \quad (24)$$

Finally the optimal control can be rewritten as

$$u = Kx + u_{ext} \quad (25)$$

where

$$K = -R^{-1}G^T P \quad (26)$$

and

$$u_{ext} = -R^{-1}G^T b_{ext} \quad (27)$$

which represents the external control signal Fig. 1 [23].

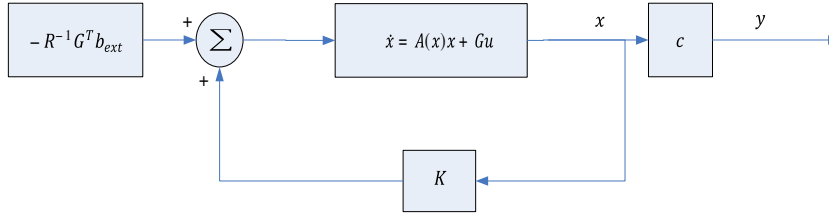


Fig. 1. Optimal trajectory control [23].

To summarize, we need to integrate the modified Riccati equations for P and b_{ext} , then the optimal control signal can be obtained. The boundary conditions for the MSDRE are $P(t_f)=0$ and $b_{ext}(t_f)=0$.

4. Spacecraft attitude maneuver/tracking using MSDRE

This section provides a background about the spacecraft dynamics and kinematics modeling with proposed disturbances, and then a pseudo-linear representation for this model is adopted such that the application of the MSDRE scheme can be implemented.

4.1. Spacecraft attitude dynamics

Euler equation is used to represent the attitude dynamics of a rigid spacecraft and is given by

$$\dot{\omega} = -I^{-1}[\omega \times (I\omega + h)] + I^{-1}(T - \dot{h}) \quad (28)$$

The momentum exchange devices are represented by the angular momentum vector h while the torque T is the total torque vector exerted on the spacecraft which includes

$$T = T_{gg} + T_{aero} + T_m + T_c \quad (29)$$

T_{gg} , T_{aero} , and T_m are the gravity gradient, aerodynamic, and magnetic disturbance torques, respectively, as represented in Ref. [24].

The control torque T_c control is generated using magnetic torquers only in a momentum biased system, and is given by

$$T_c = M \times B \quad (30)$$

where M is the dipole moment generated by the magnetic torquers and B is the Earth's magnetic field. Let us assume that there is another control signal u which constructs an orthogonal set with the two vectors M and B , then

$$M = \frac{B \times u}{|B|^2} \quad (31)$$

and

$$T_c = -\frac{[B \times][B \times]}{|B|^2} u \quad (32)$$

or

$$T_c = \bar{G}u \quad (33)$$

$[B \times]$ is a skew-symmetric matrix defined by

$$[B \times] = \begin{bmatrix} 0 & -b_3 & b_2 \\ b_3 & 0 & -b_1 \\ -b_2 & b_1 & 0 \end{bmatrix} \quad (34)$$

All vectors in the previous equations are derived in the spacecraft body frame.

4.2. Kinematics

Quaternions are used for spacecraft attitude representation, so the spacecraft kinematics are given

$$\dot{q} = \frac{1}{2}\Omega(\omega)q = \frac{1}{2}\Xi(q)\omega \quad (35)$$

where

$$q = \begin{bmatrix} q_{13} \\ q_4 \end{bmatrix} \quad (36)$$

q_{13} and q_4 are the vector and scalar parts of the quaternion, respectively.

$\Omega(\omega)$ and $\Xi(q)$ are defined as

$$\Omega(\omega) = \begin{bmatrix} -[\omega \times] & \vdots & \omega \\ \vdots & \ddots & \vdots \\ -\omega^T & \vdots & 0 \end{bmatrix} \quad (37)$$

$$\Xi(q) = \begin{bmatrix} q_4 U_{3 \times 3} + [q_{13} \times] \\ \vdots \\ -q_{13}^T \end{bmatrix} \quad (38)$$

here $U_{3 \times 3}$ is unit matrix.

The quaternion four elements satisfy the following normalization constraint:

$$q^T q = q_{13}^T q_{13} + q_4^2 = 1 \quad (39)$$

4.3. Pseudo-linear modeling

Now the task is to derive a pseudo-linear system in the form of Eq. (8) such that the MSDRE control scheme can be applied.

Starting with the state vector x resulting from the augmentation of the spacecraft dynamics with the quaternion kinematics as follows:

$$x^T = [\omega \quad q]^T \quad (40)$$

The state vector x describes the spacecraft angular velocity and attitude in the body frame w.r.t the inertial frame.

Let 'us consider the first part of the attitude dynamic Eq. (28)

$$F_1 = -I^{-1}[\omega \times (I\omega + h)] \quad (41)$$

or

$$F_1 = \frac{1}{2} \frac{\partial F_1}{\partial \omega} \omega + \frac{1}{2} I^{-1} [h \times] \omega \quad (42)$$

The attitude rotation matrix is given by

$$\Pi = (q_4^2 - q_{13}^T q_{13}) U_{3 \times 3} + 2q_{13} q_{13}^T - 2q_4 [q_{13} \times] \quad (43)$$

and its derivatives w.r.t q_i are

$$\frac{\partial \Pi}{\partial q_i} = -2q_i U_{3 \times 3} + 2[q_{13} U_i^T + U_i q_{13}^T] - 2q_4 [U_i \times] \quad i = 1 \dots 3 \quad (44)$$

U_i is i th column of the unit matrix U

$$\frac{\partial \Pi}{\partial q_4} = 2q_4 U_{3 \times 3} - 2[q_{13} \times] \quad (45)$$

From the above it can be shown that

$$\Pi = \frac{1}{2} \sum_{i=1}^4 \frac{\partial \Pi}{\partial q_i} q_i \quad (46)$$

If there is a vector T_l expressed in the inertial frame, its transformation to body frame T_b is obtained by

$$T_b = \Pi T_l \quad (47)$$

or

$$T_b = \frac{1}{2} \sum_{i=1}^4 \frac{\partial \Pi}{\partial q_i} q_i T_l \quad (48)$$

in a matrix form

$$T_b = \frac{1}{2} \frac{\partial T_b}{\partial q} q \quad (49)$$

where

$$\frac{\partial T_b}{\partial q} = \left[\frac{\partial \Pi}{\partial q_1} T_l \quad \frac{\partial \Pi}{\partial q_2} T_l \quad \frac{\partial \Pi}{\partial q_3} T_l \quad \frac{\partial \Pi}{\partial q_4} T_l \right] \quad (50)$$

Applying this result to the second part of the attitude dynamic equation i.e. the torque vectors, leads to

$$T_{gg} = \frac{1}{4} \frac{\partial T_{gg}}{\partial q} q \quad (51)$$

$$T_m = \frac{1}{2} \frac{\partial T_m}{\partial q} q \quad (52)$$

and

$$T_{aero} = \frac{1}{4} \frac{\partial T_{aero}}{\partial q} q \quad (53)$$

Note the Earth's rotation effect is neglected w.r.t spacecraft velocity. Also, the quaternion kinematics can be rewritten as follows:

$$\dot{q} = \frac{1}{2} \left[\frac{\partial \dot{q}}{\partial \omega} \omega + \frac{\partial \dot{q}}{\partial q} q \right] \quad (54)$$

Now the augmented attitude dynamics and kinematics equations can be expressed in pseudo-linear form

$$\dot{x} = [\bar{A}(x) + \bar{\Gamma}(x)]x + Gu \quad (55)$$

or

$$\dot{x} = A(x)x + Gu \quad (56)$$

where

$$A(x) = [\bar{A}(x) + \bar{\Gamma}(x)] \quad (57)$$

$$\bar{A}(x) = \frac{1}{2} \begin{bmatrix} \frac{\partial F_1}{\partial \omega} & \vdots & I^{-1} \left(\frac{\partial T_{gg}}{\partial q} + \frac{\partial T_m}{\partial q} + \frac{\partial T_{aero}}{\partial q} \right) \\ \vdots & \ddots & \vdots \\ \frac{\partial \dot{q}}{\partial \omega} & \vdots & \frac{\partial \dot{q}}{\partial q} \end{bmatrix} \quad (58)$$

and

$$\bar{\Gamma}(x) = \begin{bmatrix} \frac{1}{2} I^{-1} [h \times] & \vdots & -\frac{1}{4} I^{-1} \left(\frac{\partial T_{gg}}{\partial q} + \frac{\partial T_{aero}}{\partial q} \right) \\ \vdots & \ddots & \vdots \\ 0 & \vdots & 0 \end{bmatrix} \quad (59)$$

Consider the following system without the control vector, and $\bar{\Gamma}(x)$:

$$\dot{\bar{x}} = \bar{f}(x, t) = \bar{A}(x)x \quad (60)$$

simply

$$\bar{A}(x) = \frac{1}{2} \frac{\partial \bar{f}}{\partial x} \quad (61)$$

where $\partial \bar{f} / \partial x$ is the Jacobian

$$\frac{\partial \bar{x}}{\partial x_i} = \frac{\partial \bar{f}}{\partial x_i} = \frac{\partial \bar{A}(x)}{\partial x_i} x + \bar{A}(x)_i \quad (62)$$

where $\bar{A}(x)_i$ is i th column of the matrix $\bar{A}(x)$

From the above

$$\left(\frac{\partial \bar{A}(x)}{\partial x_i} x \right) = \bar{A}(x) \quad (63)$$

Also, it can be shown that

$$\left(\frac{\partial \bar{\Gamma}(x)}{\partial x_i} x \right) = \begin{bmatrix} 0 & \vdots & -\frac{3}{4} I^{-1} \left(\frac{\partial T_{gg}}{\partial q} + \frac{\partial T_{aero}}{\partial q} \right) \\ \vdots & \ddots & \vdots \\ 0 & \vdots & 0 \end{bmatrix} \quad (64)$$

Then

$$\left(\frac{\partial A(x)}{\partial x_i} x \right) = [A(x) + \Gamma(x)] \quad (65)$$

where

$$\Gamma(x) = -\frac{1}{2} \begin{bmatrix} I^{-1} [h \times] & \vdots & I^{-1} \left(\frac{\partial T_{gg}}{\partial q} + \frac{\partial T_{aero}}{\partial q} \right) \\ \vdots & \ddots & \vdots \\ 0 & \vdots & 0 \end{bmatrix} \quad (66)$$

The reference trajectory can be obtained by

$$\omega_{ref} = \omega_{ref}^{bo} + \Pi_{ref}^{bo} \omega^{ol} \quad (67)$$

$$q_{ref} = q_{ref}^{bo} \otimes q^{ol} \quad (68)$$

where b , o , and I stand for body, orbit, and inertial frames, respectively.

Usually the reference trajectory is given in body frame w.r.t orbit frame so the previous relations are used to derive the corresponding reference trajectory in body frame w.r.t inertial frame.

Then

$$y_{ref}^T = [\omega_{ref} \quad q_{ref}]^T \quad (69)$$

Now, we can apply the MSDRE control scheme to spacecraft attitude tracking using the backward solution of

$$-\dot{P} = PA + (2A^T + \Gamma^T)P - PGR^{-1}G^T P + Q \quad (70)$$

and

$$-\dot{b}_{ext} = [(2A^T + \Gamma^T)P - PGR^{-1}G^T]b_{ext} - Qy_{ref} \quad (71)$$

The control gain K , the external control component u_{ext} , and the optimal control signal u are obtained through Eqs. (26), (27), and (25), respectively.

4.4. Stability of the MSDRE controller

Based on Lyapunov's method, stability regimes of the MSDRE can be estimated.

Choose

$$V(\tilde{x}) = \tilde{x}^T \tilde{P} \tilde{x} \quad (72)$$

as a Lyapunov function candidate, where $\tilde{P}$ is the positive definite solution of the modified Riccati equation in Eq. (15).

Rewriting

$$V = x^T P x + x^T b_{ext} + c_{ext}^T x + d_{ext} \quad (73)$$

then the time derivative becomes

$$\dot{V} = \dot{x}^T P x + x^T \dot{P} x + x^T P \dot{x} + \dot{x}^T b_{ext} + x^T \dot{b}_{ext} + c_{ext}^T \dot{x} + c_{ext}^T \dot{x} + \dot{d}_{ext} \quad (74)$$

sub., with Eq. (8) where

$$u = -R^{-1}G^T P x - R^{-1}G^T b_{ext} \quad (75)$$

and using the modified Riccati equations

$$\dot{V} = -[\eta^T R \eta + \lambda^T R^{-1} \lambda + y_{ref}^T Q y_{ref}] - x^T [Q + (A^T + \Gamma^T)P] x - [2\lambda^T \eta + x^T (A^T + \Gamma^T) b_{ext} - 2x^T Q y_{ref}] \quad (76)$$

where $\eta = R^{-1}G^T P x$ and $\lambda = G^T b_{ext}$.

Since Q and R are positive definite matrices then the first part is negative. In case of regulation problem the third part is dropped, while the second part $x^T [Q + (A^T + \Gamma^T)P] x$ can be rewritten as $x^T S x$.

Where

$$S = Q + \bar{S} \quad (77)$$

and

$$\bar{S} = (A^T + \Gamma^T)P \quad (78)$$

The elements of Q matrix can be chosen such that the diagonal elements of S matrix satisfy the following conditions:

$$S_{i,i} > 0$$

$$|S_{i,i}| > \sum_{j=1, j \neq i}^n |S_{i,j}|$$

$$|S_{i,i}| > \sum_{j=1, j \neq i}^n |S_{j,i}| \quad (79)$$

which lead to positive definiteness of S and hence asymptotic global stability of the controller. The first inequality can be rewritten in as

$$Q_i > |\bar{S}_{i,i}| > 0$$

and the three inequalities can be combined in

$$Q_i > \sqrt{|\bar{S}_{i,i}|^2 + |\bar{S}_{i,j}|^2 + |\bar{S}_{j,i}|^2} > 0 \quad (80)$$

This means that the values of Q matrix elements should be greater than the radius of a sphere in $\bar{S}_{i,i}$, $\bar{S}_{i,j}$, and $\bar{S}_{j,i}$ coordinates, respectively, with its center located at the origin.

Now let us consider the tracking problem again rewriting Eq. (76) as a combination of two terms as

$$\dot{V} = -\dot{V}_1 - \dot{V}_2 \quad (81)$$

where

$$\dot{V}_1 = y_{ref}^T Q y_{ref} + x^T [Q + (A^T + \Gamma^T)P] x - 2x^T Q y_{ref} + x^T (A^T + \Gamma^T) b_{ext} \quad (82)$$

and

$$\dot{V}_2 = \eta^T R \eta + \lambda^T R^{-1} \lambda + 2\lambda^T \eta \quad (83)$$

The weighting matrices Q and R can be simultaneously tuned such that Eqs. (82) and (83) take positive values by satisfying the following inequalities:

$$y_{ref}^T Q y_{ref} + x^T [Q + (A^T + \Gamma^T)P] x - 2x^T Q y_{ref} \geq |x^T (A^T + \Gamma^T) b_{ext}| \quad (84)$$

and

$$\eta^T R \eta + \lambda^T R^{-1} \lambda \geq |2\lambda^T \eta| \quad (85)$$

The initial values of Q and R can be obtained by

$$R_i = \frac{1}{\|T_{max_i}\|^2} \quad (86)$$

and

$$Q_i = \frac{1}{\|x_{e_{max_i}}\|^2} \quad (87)$$

where T_{max} is the maximum torque that can be generated by the actuators and $x_{e_{max}}$ is the maximum allowable error in the each state.

4.5. Effect of actuator saturation

Due to the actuator saturation the optimal dipole moment M^* resulting from the MSDRE controller should be scaled down keeping its direction as it is. From Fig. 2 the following relation can be obtained:

$$M_s^* = \frac{M_{max}}{M_i^*} M^* \quad (88)$$

where M_s^* is the scaled dipole moment, M_{max} is the maximum dipole moment of the magnetic torquer, and M_i^* is the maximum projected dipole moment on i axis. Scaling down the dipole moment vector can be interpreted also as control

signal scaling i.e.

$$u_s^* = \beta u^* \quad (89)$$

where u^* is the optimal control signal obtained from Eq. (75), β is the scaling factor due to actuator saturation, and u_s^* is the scaled control signal. Similar argument can be applied to Eq. (76) as in the previous section to check the effect of scaling down the control signal on the system stability. Since the scaling factor satisfies

$$1 \geq \beta \geq 0 \quad (90)$$

Redefine $\eta_s^* = \beta R^{-1} G^T P x$ and $\lambda_s^* = \beta G^T b_{ext}$ it can be shown that $|\dot{V}^*| > |\dot{V}_s^*|$ which means that the actuator saturation affects the convergence time i.e., the system will take a longer time to stabilize.

5. State estimator design

The well known extended Kalman filter EKF [25] is used through this work as a state estimator to provide the MSDRE controller with the necessary states required to drive the controller.

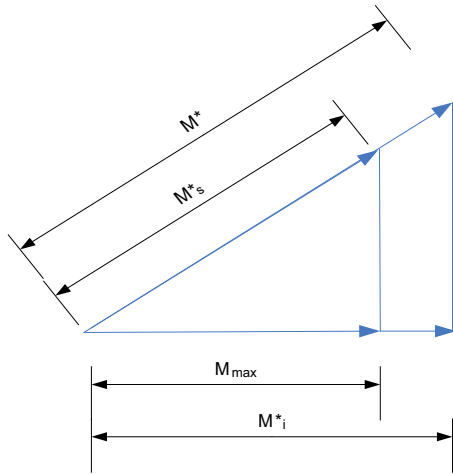


Fig. 2. Schematic diagram for scaled dipole moment.

5.1. Filter process model

The spacecraft dynamics and kinematics, given in Eqs. (28) and (35) are augmented to represent the system process model. Also, the spacecraft residual dipole and the drag coefficient are added to the state vector for more filter robustness

$$\begin{bmatrix} \dot{\omega} \\ \dots \\ \dot{q} \\ \dots \\ \dot{m} \\ \dots \\ \dot{C}_d \end{bmatrix} = \begin{bmatrix} -I^{-1}[\omega \times (I\omega + h)] + I^{-1}(T - \dot{h}) \\ \dots \\ \frac{1}{2}\Xi(q)\omega \\ \dots \\ 0 \\ \dots \\ 0 \end{bmatrix} + \begin{bmatrix} v_\omega \\ \dots \\ 0 \\ \dots \\ v_m \\ \dots \\ v_c \end{bmatrix} \quad (91)$$

$$\dot{x} = f(x, t) + v_x(t)$$

The state vector x is eleven-dimensional, defined as

$$x^T = [\omega \quad q \quad m \quad C_d]^T \quad (92)$$

5.2. Filter measurement model

The measurement equation, required for the application of the extended Kalman filtering, is derived from vector kinematics and considered to be a nonlinear function of the states according to

$$\begin{bmatrix} B_b \\ \dots \\ \dot{B}_b \end{bmatrix} = \begin{bmatrix} \Pi B_I \\ \dots \\ \Pi \dot{B}_I - \omega \times (\Pi B_I) \end{bmatrix} + \begin{bmatrix} v_B \\ \dots \\ v_{\dot{B}} \end{bmatrix} \quad (93)$$

$$z = h(x, t) + v_z(t)$$

where B_I is a vector of some reference such as the position vector to the Sun, to a star or the Earth's magnetic field vector in a reference coordinate system, and B_b is the corresponding measured vector in the spacecraft body frame. Only magnetometer measurements and its time derivatives are considered through this work. The details of the proposed filter are included in Ref. [22].

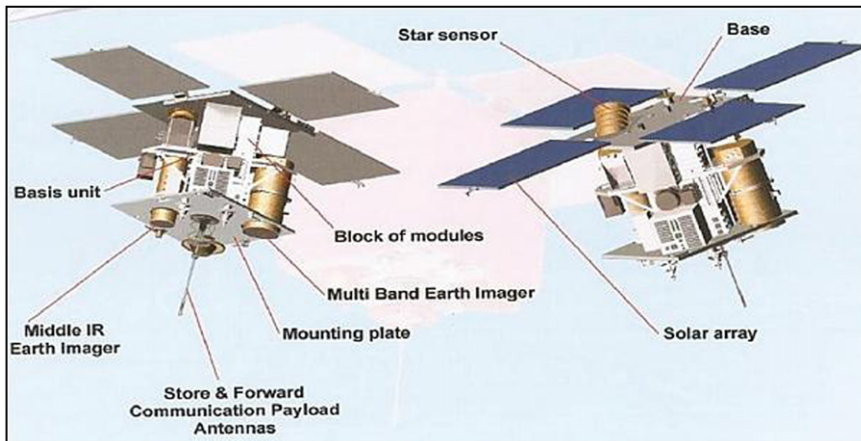


Fig. 3. EgyptSat-1 Bus.

6. Simulation results

To test the developed ADCS, EgyptSat-1, injected in sun-synchronous low Earth orbit is used as a real test case Fig. 3. The ADCS is implemented via three reaction wheels with a fourth one as a redundant, which is mounted along the pitch axis, three magnetorquers, four fiber optics gyros (FOG), three-axis magnetometer, and a star sensor. The detailed satellite hardware composition and implementation are shown in Fig. 4. The basic operation modes of the ADCS are the detumbling, standby, pre/post-imaging, and imaging modes, in addition to a safe mode. The last three modes are not covered through this paper, since it focuses only on a purely magnetic-based ADCS which is the base line design of EgyptSat-1 ADCS for the detumbling and standby modes. The initial control strategy of EgyptSat-1 was to utilize only the magnetorquers to detumble the spacecraft after separation and then to use the magnetometer reading to estimate the attitude and rates during the standby/stabilization mode. The control scheme was implemented using a PD controller in a scheduled gains fashion [26], while an EKF estimator was supposed to provide the controller with the attitude and rate information using only magnetometer measurements. Unfortunately, during the commissioning phase the reaction wheels had been used in additions to magnetorquers to detumble the spacecraft in order to maintain the power profile of the spacecraft, since it was found that using only magnetorquers would not detumble the spacecraft in the

proper time. Also, the fiber gyros onboard had to be used for rate measurements and a strap down scheme was used for obtaining the attitude. After detumbling the satellite successfully, the roll and yaw reaction wheels and the gyros were turned off, keeping the satellite in a momentum biased status via the pitch wheel. Then, the controller is switched to the standby mode utilizing only magnetorquers and the attitude and rate information from the EKF. It was noticed that the satellite autonomously switched back to detumbling mode, and the satellite failed to continue in the standby mode for several orbits. The analysis of the telemetry data showed the failure of the estimator to provide the controller with attitude and rates within the designed accuracies. So, the rate gyros turned on again and their measurements were plugged in the filter measurements which included the magnetometer data also. Then the filter retuned again and successfully provided the controller with the required information. Based on the lessons gained from the commissioning phase, the task here is to avoid using the reaction wheels and gyros and rely only on the magnetorquers and magnetometer measurements to maintain the power profile and hence the satellite life time.

The proposed controller/estimator design is utilized through both detumbling and standby/stabilization modes using only magnetorquers and magnetometer readings to meet the following ADCS requirements:

- Depress the total angular velocity after separation to less than 0.1 deg/s in one orbit and half.

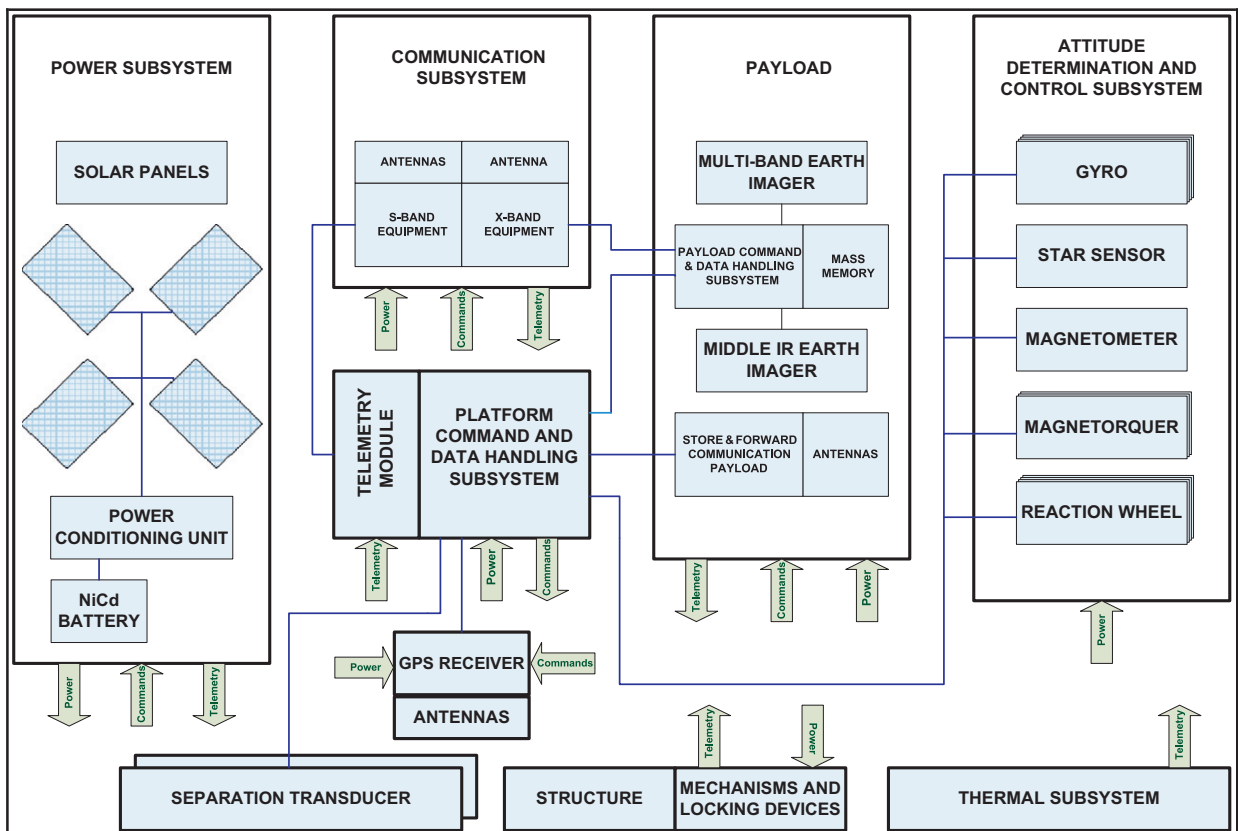


Fig. 4. EgyptSat-1 hardware layout.

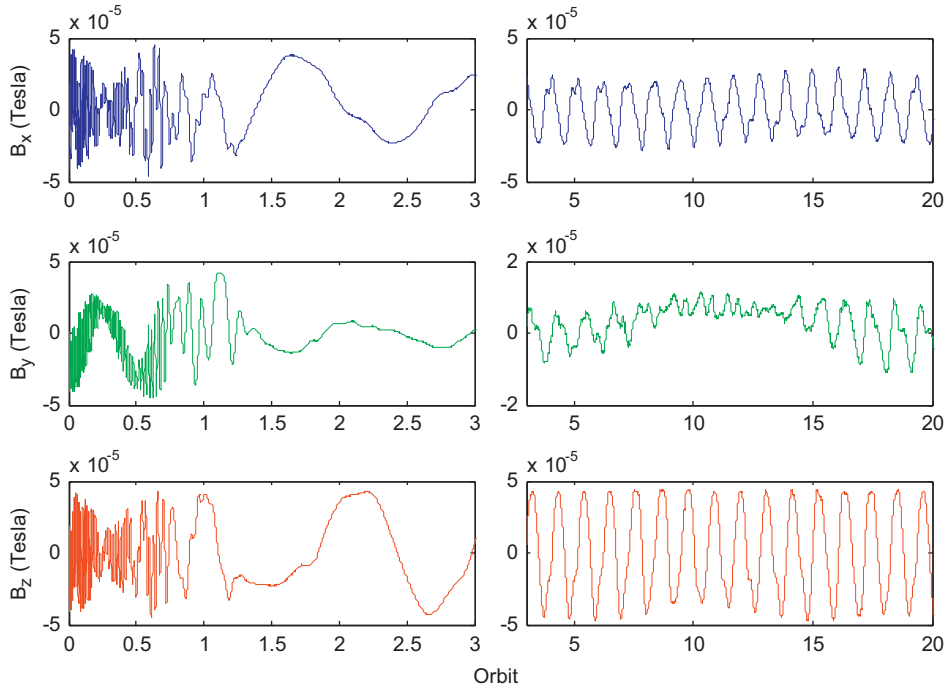


Fig. 5. Magnetometer measurements (telemetry) (detumbling/transient-standby).

- Maintain pointing accuracy during stabilization/standby mode within 5 deg and rate less than 0.05 deg/s in each axis.
- Achieve attitude determination accuracy better than 5 deg and rate of 0.03 deg/s during the same mode.

Before the discussion of the simulations results the various initial conditions used for simulation are presented as follows:

The following initial conditions are obtained from the LV provider according the worst condition of separation

- Angular velocities
 $\omega = [-4 \quad -4 \quad -2]^T \text{ deg/s}$
- Attitude in Euler angles (EA)
 $EA = [10 \quad 120 \quad 30]^T \text{ deg}$
- Pitch wheel bias
 $h = [0 \quad -0.1 \quad 0]^T \text{ N.m.s}$
- Position
 $R_S = [-1220 \quad -966.5 \quad 6854]^T \text{ km}$
- Velocity
 $V_S = [-7.426 \quad 0.1801 \quad -1.277]^T \text{ km/s}$

For the MSDRE controller, two sets of weighting matrices Q and R are used for the both flight modes,

while the switching occurs according to value of the angular velocity and scheduled according to:

- Detumbling mode

$$Q = \text{diag} \begin{bmatrix} [10^8 \quad 10^8 \quad 10^8]^T \\ [10^5 \quad 10^5 \quad 10^5 \quad 10^5]^T \end{bmatrix}$$

$$R = [10^8 \quad 10^8 \quad 10^8]^T$$

- Stabilization mode

$$Q = \text{diag} \begin{bmatrix} [2.0 \times 10^7 \quad 1.5 \times 10^8 \quad 2.0 \times 10^7]^T \\ [6.0 \times 10^6 \quad 5.0 \times 10^6 \quad 6.0 \times 10^6 \quad 10^5]^T \end{bmatrix}$$

$$R = [10^8 \quad 10^8 \quad 10^8]^T$$

- Switching condition

$$q_{1e}^{br} q_{4e}^{br} \leq 0.04 \quad \omega_x^{bo} \leq 0.0005 \text{ rad/s}$$

$$q_{3e}^{br} q_{4e}^{br} \leq 0.04 \quad \omega_z^{bo} \leq 0.0005 \text{ rad/s}$$

In addition to the magnetometer measurements, their corresponding time derivatives are computed in a semi-analytical form and included as a separate set of measurements. The details about the utilization of magnetometer measurements and their time derivatives are discussed in Ref. [22]. Fig. 5, shows the real magnetometer measurements

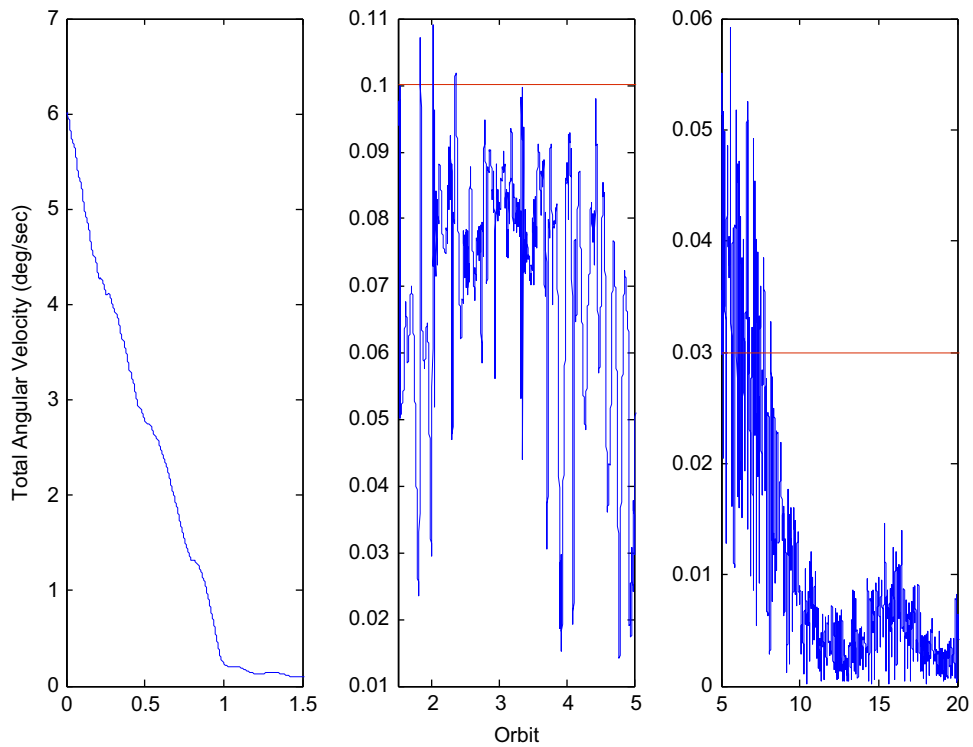


Fig. 6. Total angular velocity history (detumbling-transient-standby).

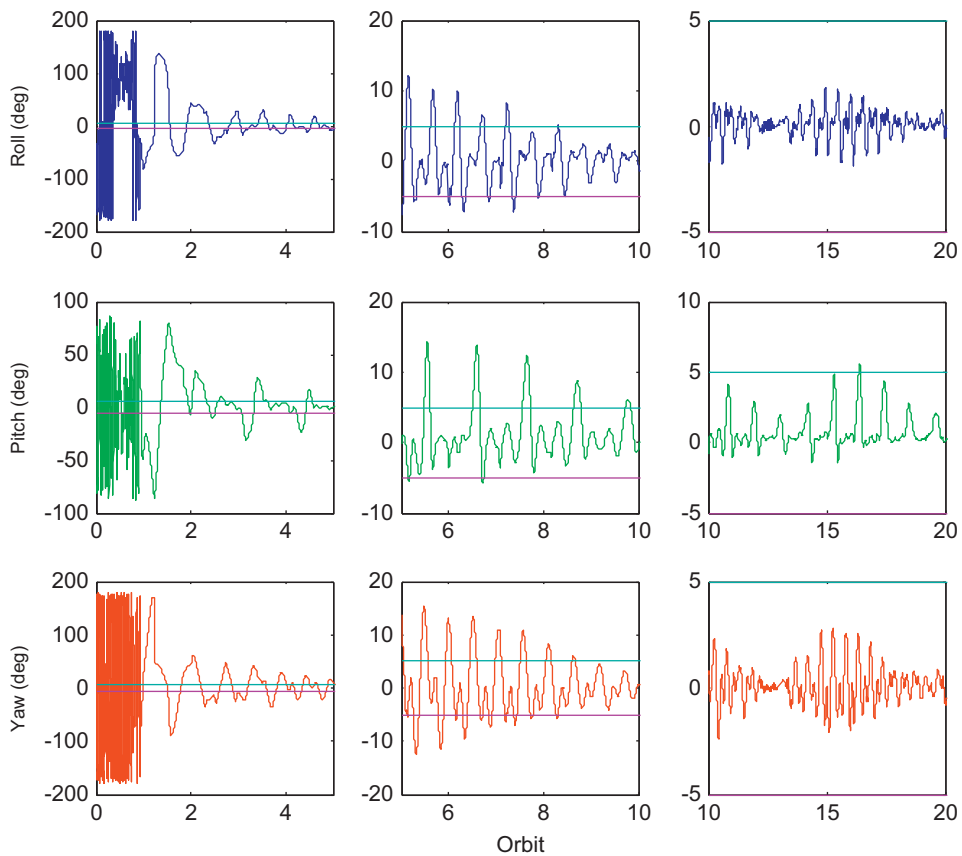


Fig. 7. Pointing error (detumbling/transient-standby).

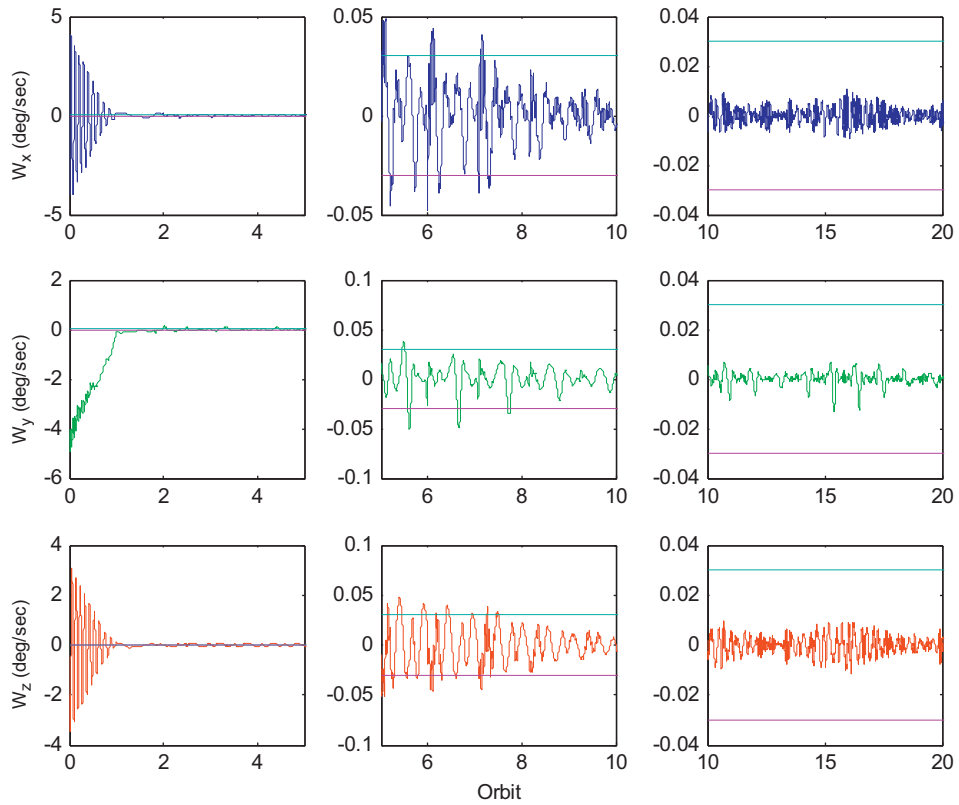


Fig. 8. Angular velocity error (detumbling/transient-standby).

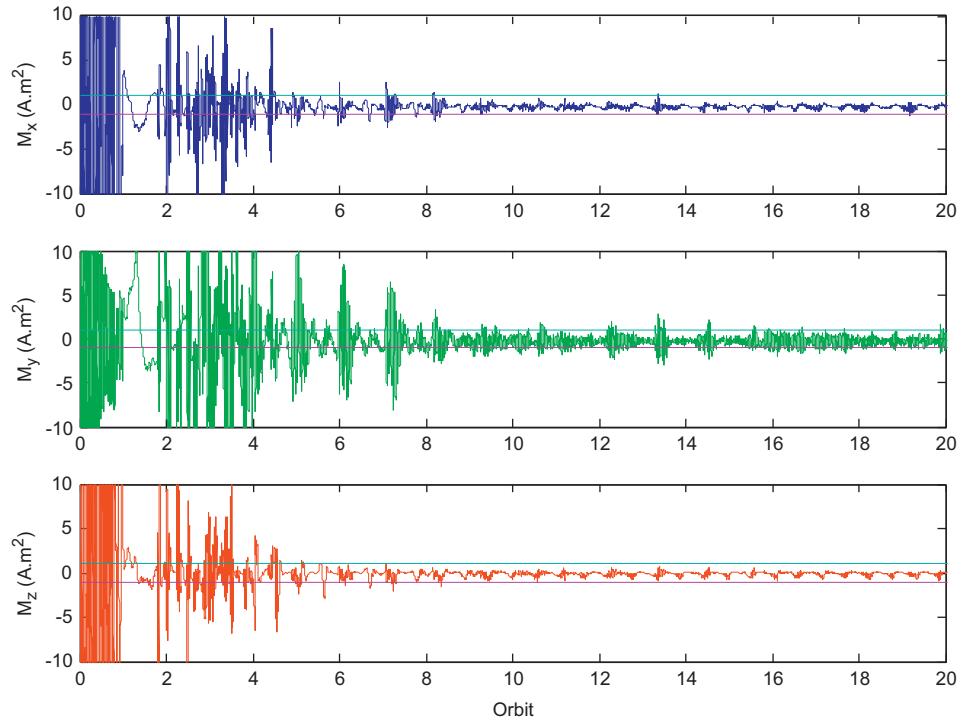


Fig. 9. Applied dipole moments.

of EgyptSat-1, extracted from the satellite telemetry. Note that, the telemetry data presented in this paper are composed of two parts. The first part obtained during the detumbling and transition to standby mode which extends up to three orbits. The second part is almost one-day flight data in a stabilized standby mode. The data obtained during the instability period mentioned previously and filter tuning is dropped. The quaternion normalization constraint in Eq. (39) is preserved using a constrained quaternion additive estimation as in Ref. [22].

The filter is initialized using initial error covariance P_{f0} and initial states x_0 defined as follows:

$$P_{f0} = \text{diag} \begin{bmatrix} [0.2094^2 & 0.2094^2 & 0.2094^2]^T & [1_{1 \times 4}]^T & [0.01^2 & 0.01^2 & 0.01^2]^T & 0.01^2 \end{bmatrix}$$

$$x_0 = [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1.0 \ 0 \ 0 \ 0 \ 0]^T$$

while the process and measurement noise covariance matrices Q_f and R_f are assembled according to

$$Q_f = \text{diag} \begin{bmatrix} [(6.25 \times 10^{-7})^2 & (5.99 \times 10^{-7})^2 & (7.04 \times 10^{-7})^2]^T & [0_{1 \times 4}]^T & [(10^{-5})^2 & (10^{-5})^2 & (10^{-5})^2]^T & 0.01^2 \end{bmatrix}$$

$$R_f = \text{diag} \begin{bmatrix} [(5.0 \times 10^{-8})^2 & (5.0 \times 10^{-8})^2 & (5.0 \times 10^{-8})^2]^T & [(2.1929 \times 10^{-7})^2 & (2.1929 \times 10^{-7})^2 & (2.1929 \times 10^{-7})^2]^T \end{bmatrix}$$

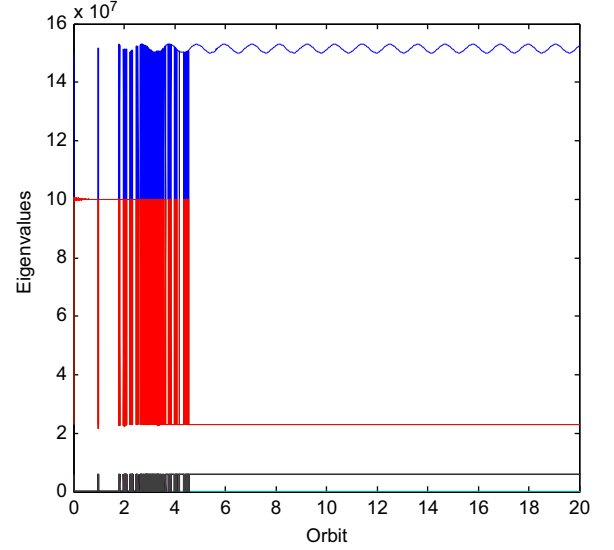


Fig. 11. S matrix eigenvalues.

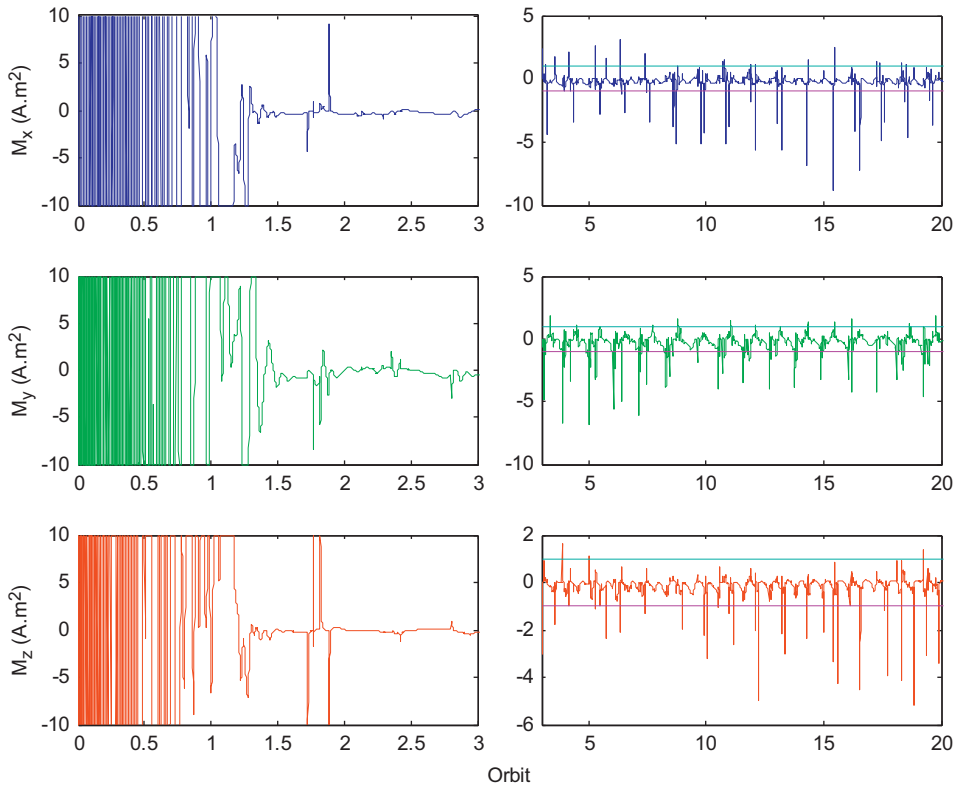


Fig. 10. Magnetorquers dipole moments (telemetry).

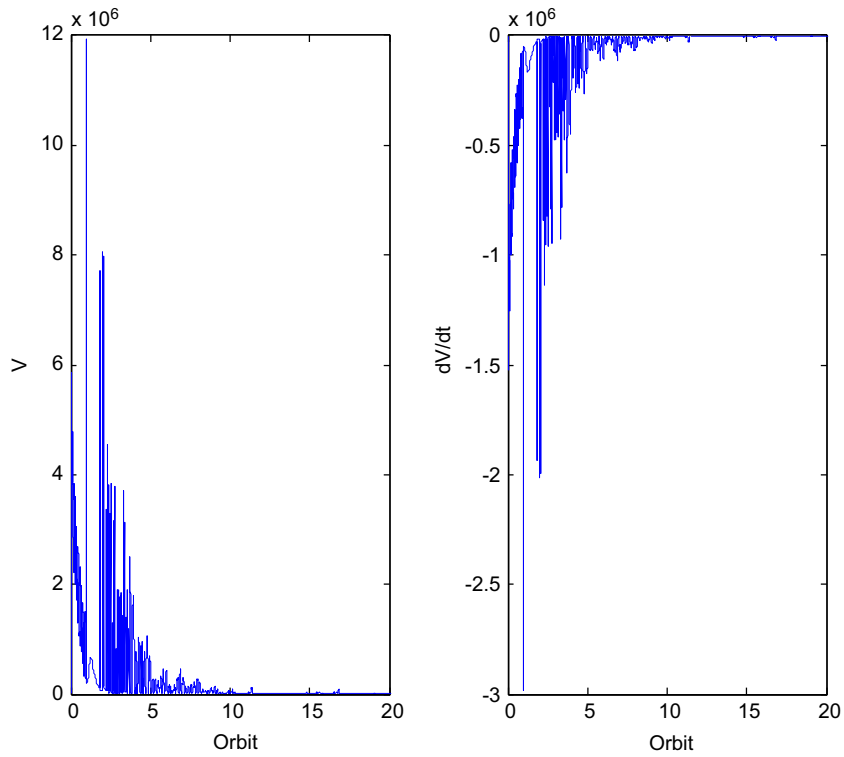


Fig. 12. Lyapunov function and its time derivative.

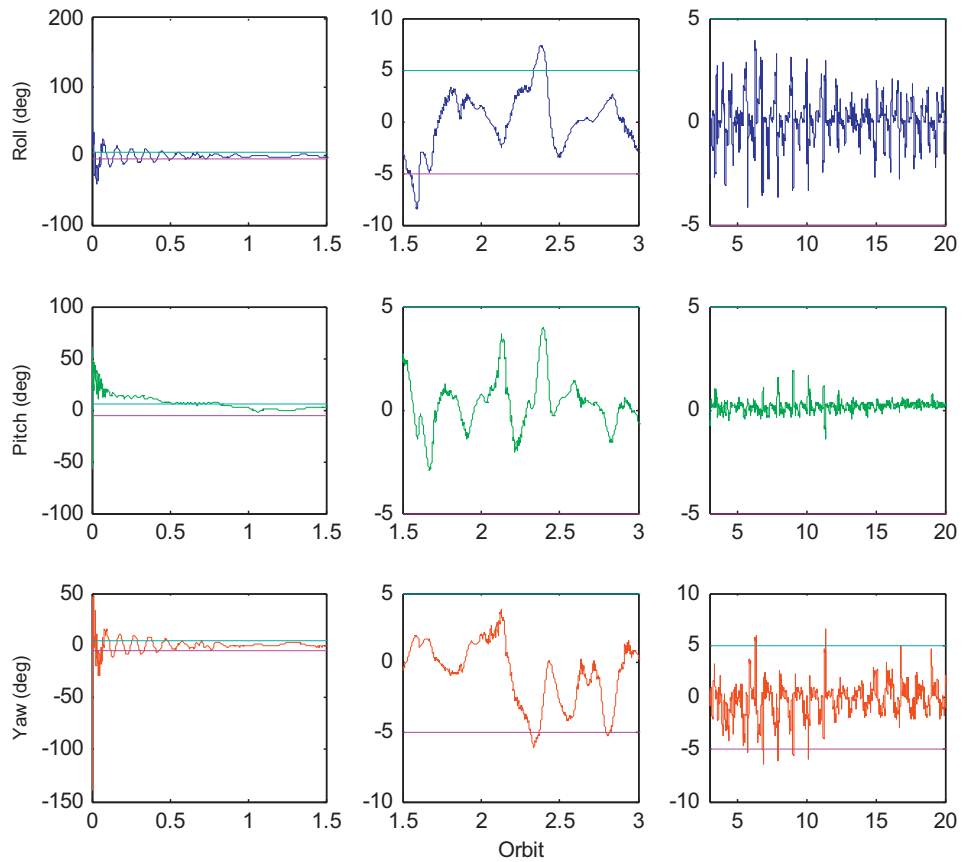


Fig. 13. Attitude estimation error (detumbling–transient–standby).

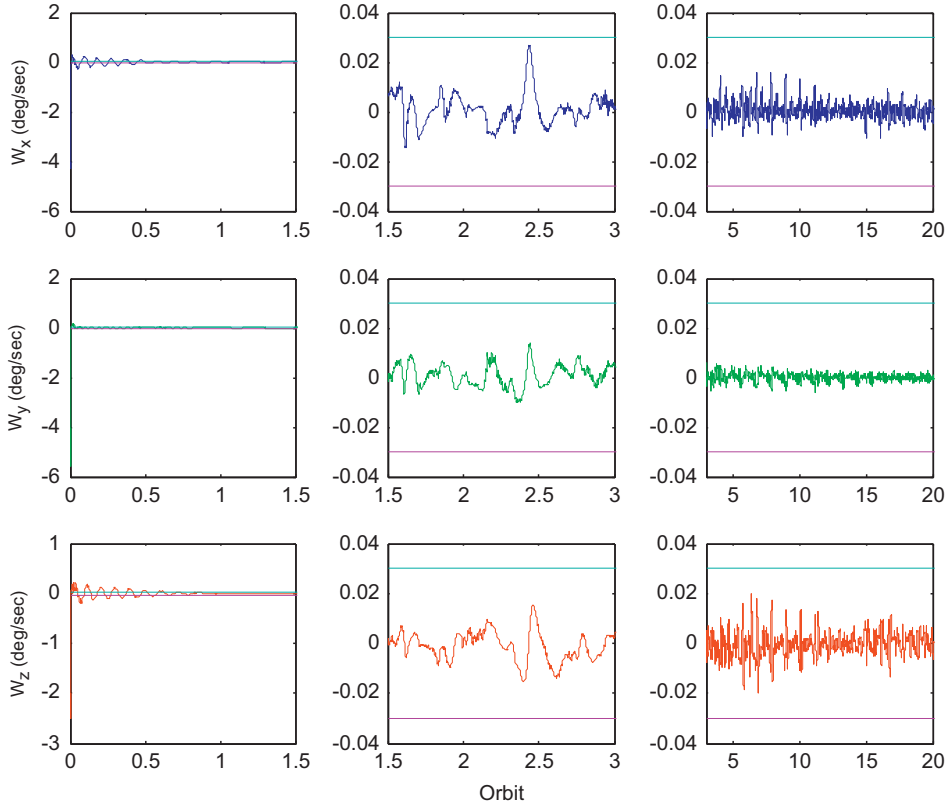


Fig. 14. Angular velocity estimation error (detumbling–transient–standby).

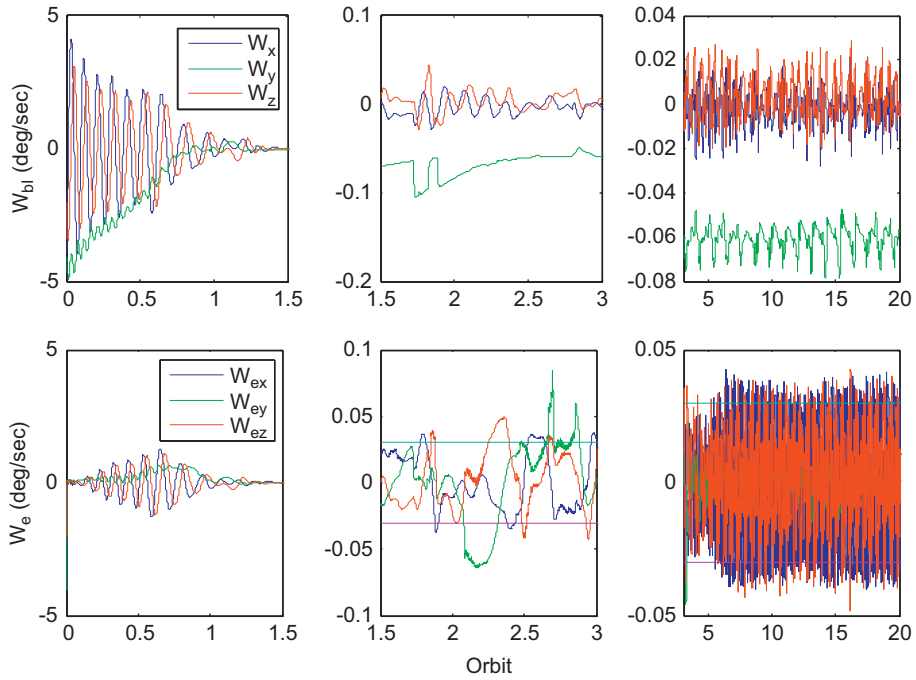


Fig. 15. Rate gyros measurements (telemetry) and filter estimation error (detumbling–transient–standby).

The rest of the simulation parameters are defined to be

- Magnetorquer strength

$$M_{max} = [10 \ 10 \ 10]^T \text{A.m}^2$$

- Spacecraft dipole

$$m = [0.3 \ 0.3 \ 0.3]^T \text{A.m}^2$$

- Drag coefficient

$$C_d = 2$$

- Time span 20 orbits
- Sampling time 4 s
- Integration algorithm used for the modified Riccati equation and state propagation is Runge–Kutta ode4
- The magnetic field vector B is based on IGRF-11 Model.

The total angular velocity is depressed after separation from 6 to 0.094 deg/s in 1.5 orbit, then continued decreasing during the transient/Earth acquisition interval which extends up to 5 orbits and maintained to less than 0.05 deg/s along the standby mode as shown in Fig. 6. The pointing and angular velocity errors are shown in Figs. 7 and 8 through three time windows. The first one represents the detumbling mode and transient and the other two windows show the performance during the standby mode. The MSDRE controller successfully drives the spacecraft from detumbling mode to an Earth pointing

mode in five orbits then the pointing error starts to be bounded within 5 deg and the angular velocity error reaches values less than 0.05 deg/s. Fig. 9 gives a picture of the history of the applied dipole moments, and Fig. 10 shows the real magnetorquers dipole moments where the first part represents their contribution during the detumbling mode including the transition to the standby mode. It can be shown that the dipole moment levels during the detumbling and transition are much lower than those resulted from the proposed controller since there is

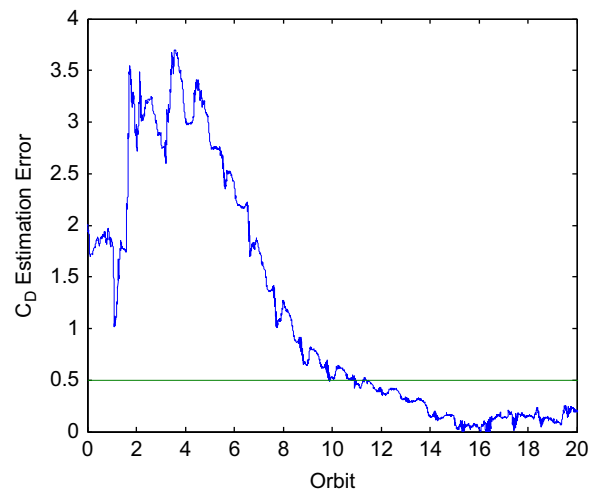


Fig. 17. Drag coefficient estimation error.

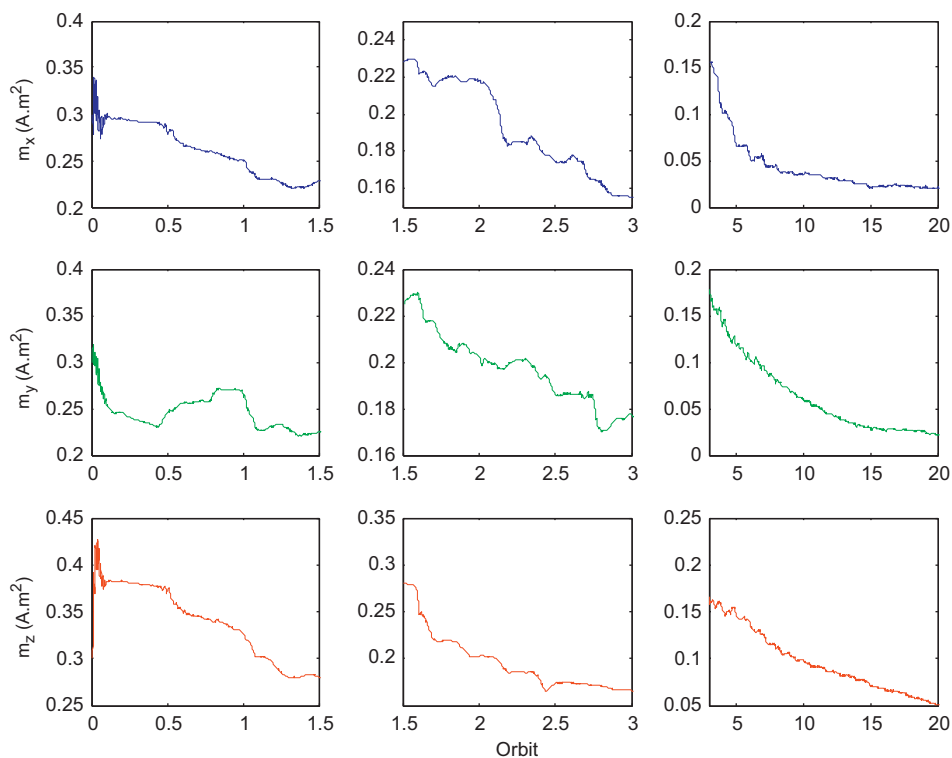


Fig. 16. Spacecraft dipole estimation error (detumbling–transient–standby).

another part of the control effort was taken over by the reaction wheels during this mode. The second part of Fig. 10 gives the real dipole moments during the standby mode where a lot of spikes can be noticed during this mode almost reaching the saturation limits, while in Fig. 9 the dipole moments are bounded to values below $\pm 1.0 \text{ A m}^2$. This is due to the fixed gains of the PD controller during this mode, while the gains of the MSDRE controllers are updated each time step, besides the scaling down technique introduced previously, which avoids the generation of unnecessary dipole moments. It should be mentioned here that the hardware configuration and sizing are kept to the original values as in EgyptSat-1.

The stability of the controller is demonstrated through Fig. 11 which shows the eigenvalues of the S matrix in Eq. (77) and Fig. 12 that indicates the history of the Lyapunov function and its time derivatives through the both modes.

Figs. 13 through 17 describe the performance of the proposed filter. The transient of the filter extends up to three orbits including the detumbling mode. Figs. 13 and 14 present the attitude and the angular velocity ω^{bo} estimation errors in the three axes. The figures show the fast convergence of the filter since it successfully estimated the attitude and rate within the required values. Fig. 15 indicates the angular velocity ω^{bl} measured by gyros and the corresponding estimation errors of the

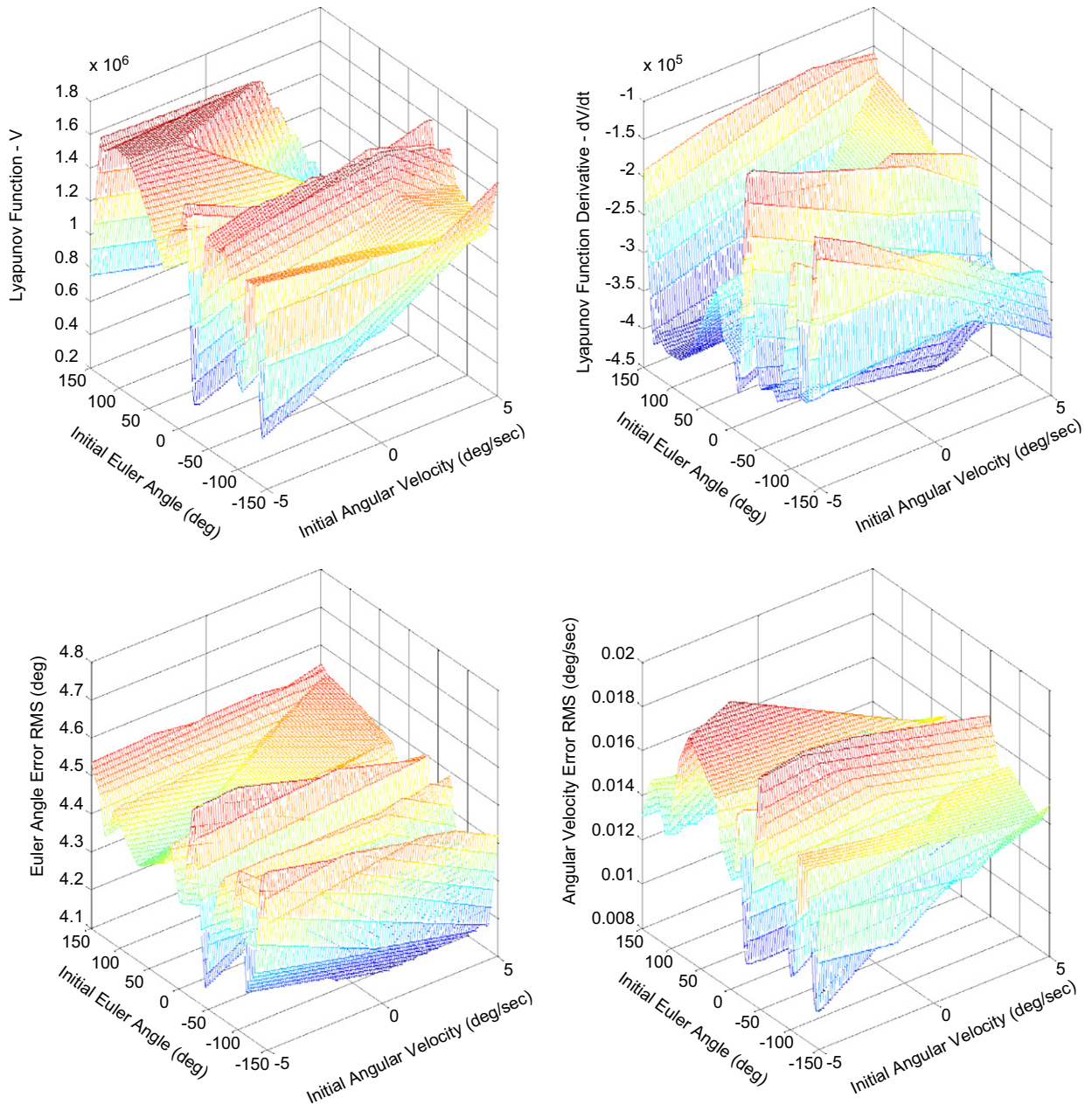


Fig. 18. MSDRE controller and EKF performance for various initial attitude and angular velocity (Monte-Carlo simulations).

filter. The filter estimation errors during the detumbling reach to high values and actually do not reflect the true performance of the filter as in Fig. 14, since the hardware utilization regarding actuation is different. During the standby mode the estimation errors are bounded closely to the designed values with spikes reach up to 0.04 deg/s.

Figs 16 and 17 indicate the estimation errors for the spacecraft dipole and the drag coefficient. For the spacecraft dipole it takes almost 14 orbits to reach estimation error less than 0.05 A m² while the drag coefficient error converges to values less than 0.5 after 10 orbits. The local observability of the system in (92) and (94) can be applied using the linearized model resulted from the EKF by computing the rank of the following observability matrix:

$$O(H, F, t_0, t_f) = \int_{t_0}^{t_f} \Phi^T(t) H^T(t) H(t) \Phi(t) dt \quad (94)$$

where $F = \partial f / \partial x$ is the Jacobian of the system.

The elements of the observability matrix can be calculated using a closed form solution of the previous integral and are given in Ref. [22]. The rank of the observability matrix was found to be 10 when the integral was performed over the sampling time $t_0 \leq t \leq t_f$ along the simulation time span. Note, although the number of estimated states is 11, the rank of the observability matrix is less by one state, this is due to the quaternion constraint in Eq. (39) which is preserved by applying the constrained quaternion additive estimation at the end of each estimation cycle. To verify the asymptotic global stability of the MSDRE controller, 300 runs of Monte-Carlo simulations are performed for a time span of 20 orbits and excluding the first five orbits that include the detumbling and transient interval. Uniformly random distributed initial conditions on the intervals $[-150150]$ deg and $[-55]$ deg/s for each axis are generated and the Lyapunov function and its time derivative are investigated. The top two sub-figures of Fig 18 represent the Monte-Carlo simulations, the x and y axes represent the average of the initial attitude and angular velocity at each run respectively, while the z -axis shows the Lyapunov function and its time derivative. Similarly, the bottom two sub-figures of Fig 18 show the stability of the filter for the same initial conditions, the z -axis indicates the Euler angle and angular velocity r.m.s errors calculated after exclusion of the first three orbits that include the detumbling and transient interval. For the controller the Lyapunov function and its derivative satisfy the stability conditions for the various initial conditions while the attitude and rate estimation accuracy are bounded to values better than 5 deg and 0.03 deg/s, respectively.

7. Conclusion

A Modified State-Dependent Riccati Equation (MSDRE) based controller has been developed and adopted for spacecraft attitude maneuver/tracking. Although, the main actuator used through this work is magnetorquers, any other type of actuators can be utilized with slight change in the control input distribution matrix. During any flight scenario, usually different control strategies are used according to the mode of operation. The developed

controller can be used for different operation modes with only scheduled weighting matrices. An extended Kalman filter (EKF) based estimator has been developed for attitude and rate estimation using only one reference sensor. While the magnetometer measurements are used through this paper, any other reference sensor can be used or added to the measurement model. The stability of the controller and estimator is demonstrated via Monte-Carlo simulations that verify the stability conditions for the Lyapunov function and indicate bounded attitude and rate estimation accuracy. Also, the concept of local observability has been applied to the estimator and verified along the simulation time span. Finally, an integrated low cost ADCS is developed for low Earth orbit spacecraft and successfully applied to a real test case and met the mission requirements.

Acknowledgments

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